Introduction to Python

Variables and Data Types

Java

In Java variables are statically typed. Which means we need to mention the data type while declaring a variable.

```
public class DeclaringVariable{
        public static void main(String[] args){
                int a = 123;
                float b = 1.23f;
                double c = 1.23;
                String d = "Hello World!";
                boolean e = true;
                boolean f = false;
        }
}
```

Python

Variables are dynammically typed. No need to write data type while declaring a variable.

```
a = 123 #integer
b = 1.23 #float
c = 1 + 2j #complex
d = "Hello World" #string
e = True #boolean true
f = False #boolean false
```

Printing the variables

Java

```
public class PrintingVariable{
        public static void main(String[] args){
                System.out.println(a);
                System.out.println(b);
                System.out.println(c);
                System.out.println(d);
                System.out.println(e);
                System.out.println(f);
        }
}
```

```
# Output
123
1.23
1.23
Hello World
true
false
```

Python

```
print (a)
print (b)
print (c) #Bracket for complex number
print (d)
print (e)
print (f)
```

```
# Output
```

```
123
1.23
(1+2j)
```



```
Hello World
True
False
```

Checking types

Java

You cannot check primitive types at runtime because java is statically typed. However for non-primitive data types, you can use `instanceof` .

```
public class CheckType{
    public static void main(String[] args){
        Object a = 123;
        System.out.println(a instanceof Integer);
    }
}
```

```
# Output
true
```

Python

```
c = 1 + 2j

type(c)
```

```
# Output
complex
```

Basic arithmetic operations

Java

For floor division both numbers should be of int datatype.

```
public class PrintingVariable{
        public static void main(String[] args){
                System.out.println(5 + 2);
                System.out.println(5 - 2);
                System.out.println(5 * 2);
                System.out.println(5.0 / 2);
                System.out.println(5 / 2);
                System.out.println(5 % 2);
                System.out.println(Math.pow(2,4));
        }
}
```

```
# Output
7
3
10
2.5
2
1
16
```

Python

```
print (5 + 2)
print (5 - 2)
print (5 * 2)
```

```
print (5 / 2)
print(5//2)
print (5 % 2)
print (2 ** 4)
```



```
# Output
7
3
10
2.5
2
1
16
```

Concatination

Java

```
public class Contatination{
        public static void main(String[] args){
                int a = 123;
                int b = 456;
                                                System.out.println("The current value of a is "+ a);
                                                System.out.println("The current value of a is " + a +
" and b is" + b);
        }
}
```

```
# Output
The current value of a is 123
The current value of a is 123 and b is 456
```

Python

```
a = 123
b = 456

# Concatenated using comma
```



```
print ("The current value of a is", a)
print ("The current value of a is", a, "and b is", b)

#Concatenated using +
print ("The current value of a is " + a)
print ("The current value of a is " + a + " and b is " + b)


# Output
The current value of a is 123
The current value of a is 123 and b is 456
The current value of a is 123
The current value of a is 123 and b is 456
```

String

String Concatination

Java

```
public class StrContatination{
        public static void main(String[] args){
                String a = "Welcome to ";
                String b = "CSE330 Lab.";
                System.out.println(a + b);
        }
```

```
}
```

```
# Output
Welcome to CSE330 Lab.
```

Python


```
a = "Welcome to "
b = "CSE330 Lab."
z = a + b
print (z)
```

```
# Output
Welcome to CSE330 Lab.
```

String Indexing

Java

```
//Accessing String with character position number. Space is a lso a character
public class StrIndexing{
        public static void main(String[] args){
                String a = "Welcome to ";
                String b = "CSE330 Lab.";
                System.out.println(a.charAt(0));
                System.out.println(b.charAt(5));
        }
}
```

```
# Output
```

```
W
0
```

Python

```
a = "Welcome to "
b = "CSE330 Lab."
```



```
print (a[0])
print (b[5])

# Output
W
0
```

Length of the String

Java

```
public class StrLength{
 public static void main(String[] args) {
 String a = "Welcome to ";
 String b = "CSE330 Lab.";
 String z = a + b;

 System.out.println(z.length());
 }
}

#Output
22
```

Python

```
a = "Welcome to "
b = "CSE330 Lab."
z = a + b
print(len(z))
```



```
#Output
22
```

Acsessing Last Character

Java

```
public class LastChar{
 public static void main(String[] args) {

 String a = "Welcome to ";
 String b = "CSE330 Lab.";

 System.out.println(b.charAt(b.length() - 1));
 }
}
```

```
#Output
 .
```

Python

```
a = "Welcome to "
b = "CSE330 Lab."
print(b[len(b)-1])
print(b[-1])
```

```
#Output
.
.
```


String Slicing

Java

```
public class StrSlice {
public static void main(String[] args) {

String s = "Welcome to CSE330 Lab.";

// using builtin function
// s[0:5] → substring starting from 0 to 5 (excludin g)
System.out.println(s.substring(0, 5));

// without using any builtin function
for(int i=0; i<5; i++){
System.out.print(s.charAt(i));
}
}
}
```

```
#Output
```

```
Welco
Welco
```

Python

```python
s = "Welcome to CSE330 Lab."
print(s[0:5]) # prints characters from index 0 to 4


#Output
Welco
Welco
```


Java

```java
public class StrSlice {
 public static void main(String[] args) {

 String s = "Welcome to CSE330 Lab.";

 // s[:6] → substring starting from 0 to 6 (excludin g).
 System.out.println(s.substring(0, 6));

 // s[6:] → substring starting from 6 to End.
 System.out.println(s.substring(6));

 // s[-4:] → last 4 characters
 System.out.println(s.substring(s.length() - 4));

 //============================================

 // without using builtin functions
 for(int i=0; i<6; i++){
 System.out.print(s.charAt(i));
```

```
}
System.out.println();
for(int i=6; i<s.length(); i++){
System.out.print(s.charAt(i));
}
System.out.println();
for(int i=s.length()-4; i<s.length(); i++){
System.out.print(s.charAt(i));
}
}
}
```



```
#Output
Welcom
e to CSE330 Lab.
Lab.
#=================
Welcom
e to CSE330 Lab.
Lab.
```

Python

```
s = "Welcome to CSE330 Lab."

print (s[:6])
print (s[6:])
print (s[-4:])
```

```
#Output
Welcom
```

e to CSE330 Lab.
Lab.

User Input

Java

```
import java.util.Scanner;

public class UserInput {
 public static void main(String[] args) {
 Scanner sc = new Scanner(System.in);
 System.out.print("Enter username: ");
 String username = sc.nextLine();
```



```
 System.out.println("Username is: " + username);
 //=======================================
 System.out.println(username + username + username);
//=======================================
 System.out.print("Enter number: ");
 int num = input.nextInt();
 System.out.println(num * 2);
 }
}
```

```
#Output
Enter username:coder
Username is: coder
#=======================================
codercodercoder
#=======================================
Enter number:4
8
```

Python

```
username = input("Enter username:")
print("Username is: " + username)
#=======================================
print(username*3)
#=======================================
num = int(input("Enter number:"))
print (num * 2)


#Output
Enter username:coder
Username is: coder
#=======================================
codercodercoder
```



```
#=======================================
Enter number:4
8
```

Conditional Statement

Java

```
public class CondStat {
public static void main(String[] args) {
int x = 5;
int y = 6;

if (x > y) {
System.out.println("Maximum is x");
} else {
System.out.println("Maximum is y");
```

```
}
}
}
```

```
#Output
Maximum is y
```

Python

```
x = 5
y = 6

if x > y:
 print("Maximum is x")
else:
 print ("Maximum is y")
```



```
#Output
Maximum is y
```

elif

Java

```
// elif
public class CondStat {
 public static void main(String[] args) {
 int x = 5;
 int y = 5;

 if (x > y) {
 System.out.println("Maximum is x");
 } else if (x == y) {
```

```
System.out.println("x and y are equal");
} else {
System.out.println("Maximum is y");
}
}
}
```

```
#Output
x and y are equal
```

Python

```
#elif
x = 5
y = 5

if x > y:
```



```
 print("Maximum is x")
elif x == y:
 print ("x and y are equal")
else:
 print ("Maximum is y")
```

```
#Output
x and y are equal
```

and

Both conditions need to be True

Java

```
public class CondStat {
 public static void main(String[] args) {
 int a = 200;
 int b = 10;
 int c = 500;

 if (a > b && c > a) {
 System.out.println("Both conditions are True");  } else {
 System.out.println("Negative!!");
 }
 }
}
```

```
#Output
Both conditions are True
```

Python


```
a = 200
b = 10
c = 500
if a > b and c > a:
 print("Both conditions are True")
else:
 print("Negative!!")
```

```
#Output
Both conditions are True
```

or

Either one of the conditions = True should be enough

Java

```
public class CondStat {
public static void main(String[] args) {
int a = 200;
int b = 10;
int c = 500;

if (a > b || a > c) {
System.out.println("At least one of the condition s is True");
} else {
System.out.println("Negative!!");
}
}
}
```



```
#Output
At least one of the conditions is True
```

Python

```
a = 200
b = 10
c = 500
if a > b or a > c:
 print("At least one of the conditions is True")
else:
 print("Negative!!")
```

```
#Output
At least one of the conditions is True
```

Nested if

Java

```
public class CondStat {
public static void main(String[] args) {
int x = 50;

if (x > 10) {
System.out.println("Above ten!");

if (x > 20) {
System.out.println("and also above 20!");
} else {
System.out.println("but not above 20!");
}
```



```
} else {
System.out.println("Not even above ten!");
}
}
}
```

```
#Output
Above ten!
and also above 20!
```

Python

```python
x = 50

if x > 10:
 print("Above ten!")
 if x > 20:
 print("and also above 20!")
 else:
 print("but not above 20!")
else:
 print("Not even above ten!")


#Output
Above ten!
and also above 20!
```

pass

Java

```java
public class PassStat{
 public static void main(String[] args) {
```



```java
 int a = 10;
 int b = 20;

 if (b > a) {
 // do nothing (equivalent to pass)
 } else {
 System.out.println("hello");
 }
 }
}
```

Python

```
a = 10
b = 20

if b > a:
 pass
#else:
#print("hello")
```

Python Built-in Data Type

List

Similar to Java or C array, but not same

List can store any type of data in one single list.

It doesn’t have any fixed size like array.

Requires more memory to store additional information about data types of each element

Python


```
#Initializing
list = [1, 2, 3, 4, 5]
print(list)


#Output
 [1, 2, 3, 4, 5]
```

```
mixed_list = [1, 2.5, "hello", [1, 2, 3]]
print(mixed_list[3])


#Output
 [1, 2, 3]


#List Methods
print(len(list))
print(4 in list) #Search

list[2] = "New" #Update value
print(list)

list.append("World") #Add new element
print(list)


#output
5
True
 [1, 2, 'New', 4, 5]
 [1, 2, 'New', 4, 5, 'World']


list.remove("New") #Remove element
print(list)
```



```
list.pop(1) #Remove using index
print(list)

list2 = [16, 55]
joined_list = list + list2
print(joined_list)
```

```
#Output
 [1, 2, 4, 5, 'World']
 [1, 4, 5, 'World']
 [1, 4, 5, 'World', 16, 55]
```

List Slicing

```
my_list = ['a', 'b', 'c', 'e', 'f', 'd']

print (my_list[3])
print (my_list[1:3])
print (my_list[:3])
print (my_list[3:])
print (my_list[-3:])


#Output
e
 ['b', 'c']
 ['a', 'b', 'c']
 ['e', 'f', 'd']
 ['e', 'f', 'd']


print (my_list[0:4:2]) #index+2 everytime
print (my_list[:]) #All elements
```



```
print (my_list[::2])
print (my_list[::-1]) #Reverse order


#Output
```

```
['a', 'c']
['a', 'b', 'c', 'e', 'f', 'd']
['a', 'c', 'f']
['d', 'f', 'e', 'c', 'b', 'a']


#Print last 4 elements in reverse
my_list = ['a', 'b', 'c', 'd', 'e', 'f']

print (my_list[-1:-5:-1])

#print 2nd to 5th element reverse order
print (my_list[4:0:-1])


#Output
['f', 'e', 'd', 'c']
['e', 'd', 'c', 'b']


#Matrix [List of list] using list
matrix = [[1, 2, 3, 4],
 [5, 6, 7, 8]]

# |1 2 3 4|
# |5 6 7 8|

print (matrix[1][2])


#Output
7
```


Tuple

Immutable (unchangable) - Items can not be modified

Collection of elements of multiple data types

Commonly used in programming (like Python) and databases to store a fixed sequence of related data

Python

```
tuple = (1, 2, 3, 4, 5)
tuple[0]
#===================
tuple[1] = "hello"


#Output
1
#======================
TypeError Traceback (most rec ent call last)
<ipython-input-41-f3de6cd25f9c> in <cell line: 0>()
----> 1 tuple[1] = "hello"

TypeError: 'tuple' object does not support item assignment
```

Set

Immutable - Items can not be modified

Unordered - No index

No duplicates allowed

Python

```
set = {"apple", "banana", "cherry", "banana"}
print(set)
```



```
#====================
set.add("orange")
```

```
print(set)

set.remove("banana")
print(set)


#Output
{'apple', 'cherry', 'banana'}
#====================
{'apple', 'cherry', 'orange', 'banana'}
{'apple', 'cherry', 'orange'}
```

Dictionary

Values stored as key : value pair

Unordered, mutable

Key must be unique and immutable types (e.g., strings, numbers, tuples).

Python

```
drivers = {
 16: "Charles",
 1: "Max",
 55: "Carlos",
 "Extra": "Liam"
}

print(drivers)


#Output
```



```
{16: 'Charles', 1: 'Max', 55: 'Carlos', 'Extra': 'Liam'}
```

```
print(drivers["Extra"]) #Accessing elements

drivers["Extra"] = "Ollie"
print(drivers)

drivers.update({81:"Oscar"})
print(drivers)

drivers.pop("Extra")
print(drivers)


#Output
Liam
{16: 'Charles', 1: 'Max', 55: 'Carlos', 'Extra': 'Ollie'}
{16: 'Charles', 1: 'Max', 55: 'Carlos', 'Extra': 'Ollie', 81: 'Oscar'}
{16: 'Charles', 1: 'Max', 55: 'Carlos', 81: 'Oscar'}
```

Loops

While loop

Java

```
public class WhileLoop{
public static void main(String[] args) {
int i = 0;

while (i < 10) {
System.out.println(i);
i = i + 1;
}
```



```
}
```

```
}
```

Python

```
i = 0
while i < 10:
 print(i)
 i = i + 1


#Output
0
1
2
3
4
5
6
7
8
9
```

For loop

Java

```
public class ForLoop{
 public static void main(String[] args) {
 for (int i = 0; i < 10; i += 2) {
        System.out.println(i);
 }
 }
}
```

Python

```
for i in range(0, 10, 2):
 print(i)


#Output
0
2
4
6
8
```

Iterate over the list

Java

```
//Iterate over the list
public class ArrayIteration{
        public static void main(String[] args){
                String[] list = {"a","b","c","d","e"};
                for(int i = 0; i < list.length; i++){
                        System.out.println(list[i]);
                }
                System.out.println("------------------");
                for(int i = 0; i < list.length; i++){
                        System.out.println(i);
                }
                System.out.println("------------------");
                for(int i = 0; i < list.length; i++){
                                                System.out.println("Index = "+i+", Value = "+list
[i]);
                }
```

```
        }
}
```

```
#Output
a
b
c
d
e
------------------
0
1
2
3
4
------------------
Index = 0, Value = a
Index = 1, Value = b
Index = 2, Value = c
Index = 3, Value = d
Index = 4, Value = e
```

Python

```
#Iterate over the list
list = ['a', 'b', 'c', 'd', 'e']

for i in list:
 print (i)
print("------------------")
for i in range(len(list)):
```

```
 print(i)
print("------------------")
```



```
for i in range(0, len(list)):
 print("Index =", i, "Value =", list[i])


# OUTPUT
a
b
c
d
e
------------------
0
1
2
3
4
------------------
Index = 0, Value = a
Index = 1, Value = b
Index = 2, Value = c
Index = 3, Value = d
Index = 4, Value = e
```

Break Statement

Java

Break statement - Out of the loop

```
public class BreakStatement{
      public static void main(String[] args){
```

```
                String[] fruits = {"apple", "banana", "cherry"};
        for(int i = 0; i < fruits.length; i++){
            System.out.println(fruits[i]);
            if(fruits[i].equals("banana")){
                break;
```



```
            }
        }
    }
}
```

```
# OUTPUT
apple
banana
```

Python

```
fruits = ["apple", "banana", "cherry"]
for x in fruits:
 print(x)
 if x == "banana":
 break
```

```
# OUTPUT
# OUTPUT
apple
banana
```

Continue Statement

Java

Continue statement - Stop the current iteration of the loop and continue with the next

```
public class ContinueStatement{
    public static void main(String[] args){
                        String[] fruits = {"apple", "banana", "cherry"};
        for(int i = 0; i < fruits.length; i++){
            if(fruits[i].equals("banana")){
```



```
                continue;
            }
            System.out.println(fruits[i]);
        }
    }
}
```

```
# OUTPUT
apple
cherry
```

Python

```
fruits = ["apple", "banana", "cherry"]
for i in fruits:
 if i == "banana":
 continue
 print(i)
```

```
# OUTPUT
apple
cherry
```

Nested Loop

Java

```
public class NestedLoop{
    public static void main(String[] args){
        int[] num_arr = {1,2,3};
        String[] alphabet_arr = {"a","b","c"};
        for(int i = 0; i < num_arr.length; i++){
            System.out.println(num_arr[i]);
```



```
                for(int j = 0; j < alphabet_arr.length; j++){
                System.out.println(alphabet_arr[j]);
            }
        }
    }
}
```

```
# OUTPUT
1
a
b
c
2
a
b
c
3
a
b
c
```

Python

```
#Nested loop
```

```
num_list = [1 , 2, 3]
alphabet_list = ['a', 'b', 'c']

for number in num_list:
 print(number)
 for letter in alphabet_list:
 print(letter)


# OUTPUT
1
```



```
a
b
c
2
a
b
c
3
a
b
c
```

Functions or Methods

Java

We cannot create a java method with default value but we can use method overloading (same method name but different numbers of parameters).

```
public class JavaMethod{
    public static void hello_method(){
        System.out.println("Hello");
    }
```

```
                        public static void name_method(String fname, String lnam
e){
            System.out.println(fname+" "+lname);
        }
        public static int sum_numbers(int num1, int num2){
            return num1 + num2;
        }
        public static int increment(int number) {
return increment(number, 1); // default value
        }
        public static int increment(int number, int by) {
return number + 2 * by;
```



```
        }
        public static void main(String[] args){
            hello_method();
            name_method("Cristiano", "Ronaldo");
            int x = sum_numbers(2, 3);
            System.out.println(x);
            System.out.println(increment(2,5));
            System.out.println(increment(2));
        }
}
```

```
# OUTPUT
Hello
Cristiano Ronaldo
5
4
12
```

Python

```
def hello_function():
 print("Hello")

def name_function(fname, lname):
 print(fname, " ", lname)

def sum_numbers(number1, number2):
 result = number1+number2
 return result

def increment(number, by=1):
 return number + 2*by

hello_function()
```



```
name_function("Cristiano", "Ronaldo")
x = sum_numbers(2, 3)
print(x)
print(increment(2))
print(increment(2, 5))
# Keyword arguments
print(increment(by=1, number=2))
```

NumPy

Java

```
import java.util.Arrays;
public class JavaArr{
    public static void main(String[] args){
                                   String[] arr = {"1", "2", "3", "2.45", "hello"};
        System.out.println(Arrays.toString(arr));
                                System.out.println("Arr Length: " + arr.length);
```

```
    }
}
```

```
# OUTPUT
 [1, 2, 3, 2.45, hello]
Arr Length: 5
```

Python

A Python library

Used for working with arrays

Short form of "Numerical Python”

Aims to provide an array object that is up to 50x faster than traditional Python lists

Provides a lot of supporting functions


```
# As arrays can store only one type of data, therefore it con verts everything to
String
# We cannot perform any kind of calculations
import numpy as np

arr = np.array([1 , 2, 3, 2.45, "hello"])
print(arr)
print(type(arr))
print("Arr Length:",len(arr))
print(arr.shape)
```

```
# OUTPUT
 ['1' '2' '3' '2.45' 'hello']
<class 'numpy.ndarray'>
5
 (5,)
```

Append

Java

```
import java.util.Arrays;
public class JavaAppend{
        public static void main(String[] args){
                                                // We need to create another array (y), where the siz
e is greater than the previous
                                                // array (x). Then copy all values from x to y and fi
nally insert the new value (70)
                int[] x = {1,2,3,4,5};
                int[] y = new int[x.length+1];
                for(int i = 0; i < x.length; i++){
                        y[i] = x[i];
                }
```



```
                y[y.length-1] = 70;
                System.out.println(Arrays.toString(y));
                System.out.println(Arrays.toString(x));
        }
}

# OUTPUT
[1,2,3,4,5,70]
[1,2,3,4,5]
```

Python

```
import numpy as np

x = np.array([1, 2, 3, 4, 5])
```

```
y = np.append(x, [70])
print(y)
print(x) #No change in main array
```

```
[ 1 2 3 4 5 70]
[1 2 3 4 5]
```

Delete

Java

```
import java.util.Arrays;

public class JavaDelete {
public static void main(String[] args) {
int[] x = {1, 2, 3, 4, 5};
// We need to create another array (y), where the siz e is less than the previous
```



```
// array (x). Then copy all values from x to
y while checking the value's index you want to remove and the n insert the other
values other than the removed value.
int deleteValue = 3; // value we want to remove

int[] y = new int[x.length - 1];

int index = 0;
for (int i = 0; i < x.length; i++) {
if (x[i] != deleteValue) {
y[index] = x[i];
index++;
}
}

System.out.println(Arrays.toString(y));
```

```
 System.out.println(Arrays.toString(x));
 }
}
```

```
#Output
 [1, 2, 4, 5]
 [1, 2, 3, 4, 5]
```

Python

```
x = np.array([1, 2, 3, 4, 5])
y = np.delete(x, 3) #Delete using index
print(x)
print(y)
```

```
#Output
 [1, 2, 4, 5]
```



```
 [1, 2, 3, 4, 5]
```

Sorting

Java

```
import java.util.Arrays;

public class SortArrayAscending {
 public static void main(String[] args) {
 int[] numbers = {5, 2, 8, 1, 9};

 Arrays.sort(numbers); //Ascending
```

```
//Descending - Reverse the Ascending order sorted arr ay
int[] revArr = new int[numbers.length];
int s = 0;
for(int i=numbers.length-1; i>=0; i--){
revArr[s++]=numbers[i];
}

System.out.println(Arrays.toString(numbers));
System.out.println(Arrays.toString(revArr));
}
}
```

```
#Output
[1, 2, 5, 8, 9]
[9, 8, 5, 2, 1]
```

Python


```
x = np.array([7, 8, 3, 10, 5])
sorted_x = np.sort(x)
print(sorted_x) #Ascending
print(sorted_x[::-1]) #Descending - Reverse the Ascending ord er sorted array
```

```
#Output
[1, 2, 5, 8, 9]
[9, 8, 5, 2, 1]
```

2-D Array

Java

```java
import java.util.Arrays;

public class Java2DArrayExample {
 public static void main(String[] args) {

 int[][] arr_2d = {
 {1, 2, 3, 7, 55},
 {4, 5, 6, 99, 10}
 };

 System.out.println(Arrays.deepToString(arr_2d));
 }
}


#Output
 [[1, 2, 3, 7, 55], [4, 5, 6, 99, 10]]
```

Python


```python
arr_2d = np.array([[1, 2, 3, 7, 55],
 [4, 5, 6, 99, 10]])
print(arr_2d)


#Output
 [[ 1 2 3 7 55]
   [ 4 5 6 99 10]]
```

Shape & Dimention of a Matrix

Java

```
public class Java2DArrayInfo {
 public static void main(String[] args) {

 int[][] arr_2d = {
 {1, 2, 3, 4, 5},
 {6, 7, 8, 9, 10}
 };

 int rows = arr_2d.length;
 System.out.println("Number of rows: " + rows);

 int cols = arr_2d[0].length;
 System.out.println("Shape: (" + rows + ", " + cols + ")");

 int ndim = 2; // all 2D arrays in Java are 2-dimensi onal
 System.out.println("Number of dimensions: " + ndim);  }
}
```



```
#Output
Number of rows: 2
Shape: (2, 5)
Number of dimensions: 2
```

Python

```
print(len(arr_2d))
print(arr_2d.shape) #row, column
print(arr_2d.ndim) #Dimension
```

```
#Output
2
(2, 5)
```

2

Indexing

Java

```
//1st & last element of an array
import java.util.Arrays;

public class SortArrayAscending {
 public static void main(String[] args) {
 int[] numbers = {5, 2, 8, 1, 9};


 System.out.println(numbers[0]);
 System.out.println(numbers[numbers.length-1]);
 }
}
```



```
#Output
5
9
```

Python

```
arr = np.array([5, 2, 8, 1, 9])

print(arr[0])
print(arr[-1])
```

```
#Output
5
```

9

Java

```
import java.util.Arrays;

public class Java2DArrayAccess {
 public static void main(String[] args) {

 int[][] arr = {
 {1, 2, 3, 4, 5},
 {6, 7, 8, 9, 10}
 };

 System.out.println(Arrays.deepToString(arr));

 System.out.println("4th element on 2nd row: " + arr [1][3]);
 }
}
```



```
[[1, 2, 3, 4, 5], [6, 7, 8, 9, 10]]
4th element on 2nd row: 9
```

Python

```
arr = np.array([[1,2,3,4,5],
 [6,7,8,9,10]])
print(arr)

print('4th element on 2nd row: ', arr[1, 3])


#Output
[[ 1 2 3 4 5]
```

```
 [ 6 7 8 9 10]]
4th element on 2nd row: 9
```

Slicing 2-D Array

Java

```
import java.util.Arrays;

public class Java2DSlicing {
public static void main(String[] args) {

int[][] arr = {
{1, 2, 3, 4, 5},
{6, 7, 8, 9, 10}
};
                                    //From the second row, slice elements from in
dex 1 to index 4 (not included)
int[] slice1 = Arrays.copyOfRange(arr[1], 1, 4);
System.out.println(Arrays.toString(slice1));
```



```
                                    //From both rows, slice index 1 to index 4 (n
ot included)
int[][] slice2 = new int[2][3];

for (int i = 0; i < 2; i++) {
slice2[i] = Arrays.copyOfRange(arr[i], 1, 4);
}

System.out.println(Arrays.deepToString(slice2));
}
}


#Output
```

```
[7, 8, 9]
[[2, 3, 4], [7, 8, 9]]
```

Python

```
arr = np.array([[1, 2, 3, 4, 5],
 [6, 7, 8, 9, 10]])

#From the second row, slice elements from index 1 to index 4 (not included)
print(arr[1, 1:4])

#From both rows, slice index 1 to index 4 (not included)
print(arr[0:2, 1:4])


#Output
[7 8 9]
[[2 3 4]
  [7 8 9]]
```


Taking last column and row

Java

```
import java.util.Arrays;

public class JavaColumnRowLoop {
 public static void main(String[] args) {

 int[][] arr = {
 {1, 2, 3, 4, 5},
 {6, 7, 8, 9, 10}
 };
```

```
int[] column0 = new int[arr.length];
for (int i = 0; i < arr.length; i++) {
column0[i] = arr[i][0];
}
System.out.println(Arrays.toString(column0));

int[] row0 = new int[arr[0].length];
for (int j = 0; j < arr[0].length; j++) {
row0[j] = arr[0][j];
}
System.out.println(Arrays.toString(row0));
}
}
```

```
#Output
[1 6]
[1 2 3 4 5]
```

Python


```
print(arr[:, 0]) #column
print(arr[0, :]) #row
```

```
#Output
[1 6]
[1 2 3 4 5]
```

Zero Matrix

Java

```java
import java.util.Arrays;

public class JavaZeros {
 public static void main(String[] args) {

 // Create a 4×2 2D array filled with zeros
 int[][] y = new int[4][2];

 System.out.println(Arrays.deepToString(y));
 }
}
```

```
#Output
 [[0, 0], [0, 0], [0, 0], [0, 0]]
```

Python

```python
#zero matrix
y = np.zeros((4,2)) #row, column
print(y)
```



```
#Output
 [[0. 0.]
   [0. 0.]
   [0. 0.]
   [0. 0.]]
```

Summation & Product of all elements

Java

```java
import java.util.Arrays;

public class JavaArraySumProd {
public static void main(String[] args) {

int[] a = {3, 4, 5};

int sumA = 0;
int prodA = 1;
for (int i = 0; i < a.length; i++) {
sumA += a[i];
prodA *= a[i];
}
System.out.println("Sum of a: " + sumA);
System.out.println("Product of a: " + prodA);
}
}
```

```
#Output
Sum of a: 12
Product of a: 60
```

Python


```python
a = np.array([3, 4, 5])

print(np.sum(a))
print(np.prod(a))
```

```
#Output
12
60
```

Column & Row Wise Operation

Java

```
import java.util.Arrays;

public class JavaArraySumProd {
public static void main(String[] args) {

int[][] b = {
{1, 2, 3},
{4, 5, 6}
};
int sumB = 0;
for (int i = 0; i < b.length; i++) {
for (int j = 0; j < b[i].length; j++) {
sumB += b[i][j];
}
}
System.out.println("Sum of b: " + sumB);

int[] colSum = new int[b[0].length];
for (int j = 0; j < b[0].length; j++) {
for (int i = 0; i < b.length; i++) {
```



```
colSum[j] += b[i][j];
}
}
System.out.println("Column-wise sum: " + Arrays.toStr ing(colSum));

int[] rowSum = new int[b.length];
for (int i = 0; i < b.length; i++) {
for (int j = 0; j < b[i].length; j++) {
```

```
rowSum[i] += b[i][j];
}
}
System.out.println("Row-wise sum: " + Arrays.toString (rowSum));
}
}
```

```
#Output
Sum of b: 21
Column-wise sum: [5, 7, 9]
Row-wise sum: [6, 15]
```

Python

```
b = np.array([[1,2,3],
 [4,5,6]])

print(np.sum(b))
print(np.sum(b, axis=0))
print(np.sum(b, axis=1))
```

```
#Output
21
```



```
[5 7 9]
[ 6 15]
```

Iterating Row by Row

Java

```
public class Java2DArrayLoopOnly {
```

```
public static void main(String[] args) {

int[][] arr = {
{1, 2, 3},
{4, 5, 6}
};

for (int i = 0; i < arr.length; i++) {
for (int j = 0; j < arr[i].length; j++) {
System.out.print(arr[i][j] + " ");
}
System.out.println();
}
}
}
```

```
#Output
1 2 3
4 5 6
```

Python

```
arr = np.array([[1, 2, 3],
 [4, 5, 6]])
```



```
for x in arr:
 print(x)
```

```
#Output
[1 2 3]
[4 5 6]
```

Print Element by Element

Java

```
public class Java2DArrayNestedLoop {
public static void main(String[] args) {

int[][] arr = {
{1, 2, 3},
{4, 5, 6}
};
for (int i = 0; i < arr.length; i++) {
for (int j = 0; j < arr[i].length; j++) {
System.out.println(arr[i][j]);
}
}
}
}
```

```
#Output
1
2
3
4
5
6
```


Python

```
arr = np.array([[1, 2, 3], [4, 5, 6]])
for x in arr:
for y in x:
```

```
print(y)
```

```
#Output
1
2
3
4
5
6
```

Linspace

Create a random array, same difference between the numbers

Java

```
public class JavaLinspace {
public static void main(String[] args) {

double start = 0;
double end = 100;
int n = 4;

double[] x = new double[n];
double step = (end - start) / (n - 1);

for (int i = 0; i < n; i++) {
x[i] = start + i * step;
}
```



```
for (int i = 0; i < n; i++) {
System.out.print(x[i] + " ");
}
}
```

```
}
```

```
#Output
0.0 33.333333333333336 66.66666666666667 100.0
```

Python

```
x = np.linspace(0, 100, 4) #start, end, how many numbers print(x)
```

```
#Output
 [ 0. 33.33333333 66.66666667 100. ]
```